

Activity

Group members:

Grad applications data

Using the grad applications data, you fit a model with GPA as the explanatory variables. Here is the regression output:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-4.3576	1.0353	-4.209	2.57e-05 ***
gpa	1.0511	0.2989	3.517	0.000437 ***

And here is the variance-covariance matrix for the fitted model:

	(Intercept)	gpa
(Intercept)	1.0718814	-0.3076994
gpa	-0.3076994	0.0893229

Questions

1. Calculate the estimated log odds of admission for a student with GPA 3.8.

$$\begin{aligned}\widehat{\log \text{odds}} &= -4.3576 + 1.0511(3.8) \\ &= \boxed{-0.36342}\end{aligned}$$

Fact: For two variables U , V and constants a, b :

$$\text{Var}(aU + bV) = a^2\text{Var}(U) + b^2\text{Var}(V) + 2ab\text{Cov}(U, V)$$

For a student with GPA 3.8, the estimated log odds is

$$\widehat{\log \text{odds}} = \hat{\beta}_0 + \hat{\beta}_1(3.8)$$

2. Using the fact above, and the variance-covariance matrix from the R output, calculate $Var(\widehat{\log \text{ odds}})$ when GPA = 3.8.

$$\begin{aligned} Var(\widehat{\log \text{ odds}}) &= Var(\hat{\beta}_0) + (3.8)^2 Var(\hat{\beta}_1) + 2(3.8) Cov(\hat{\beta}_0, \hat{\beta}_1) \\ &= 1.0719 + (3.8)^2 (0.0893) + 2(3.8) (-0.3077) \\ &= \boxed{0.0229} \end{aligned}$$

3. From your answer in question 2, calculate $SE(\widehat{\log \text{ odds}})$.

$$\begin{aligned} SE(\widehat{\log \text{ odds}}) &= \sqrt{Var(\widehat{\log \text{ odds}})} \\ &= \sqrt{0.0229} = \boxed{0.1513} \end{aligned}$$

A 95% CI for the *true* log odds of admission for students with GPA 3.8 is

$$\widehat{\log \text{ odds}} \pm 1.96 SE(\widehat{\log \text{ odds}})$$

4. Using your answers to questions 1 and 3, calculate a 95% CI for the true log odds of admission for students with GPA 3.8.

$$\begin{aligned} &-0.3634 \pm 1.96(0.1513) \\ &= \boxed{(-0.66, -0.067)} \end{aligned}$$

5. Finally, transform your confidence interval from question 4 into a 95% CI for the true *probability* of admission for students with GPA 3.8.

$$\frac{e^{-0.66}}{1 + e^{-0.66}} = 0.341$$

$$\frac{e^{-0.067}}{1 + e^{-0.067}}$$

$$= 0.483$$

$$\boxed{(0.341, 0.483)}$$